Tool Calling Under Interface Drift: A Non-Oracle Study of Validation and Repair

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Abstract. Tool-calling language models rely heavily on surface-form tool interfaces, including descriptions, parameter names, and candidate tool sets. In realistic deployments, these interfaces drift after release even when the underlying tool semantics remain unchanged. We study this problem as *interface drift* and evaluate a lightweight training-free mitigation, SchemaShield-Lite, built from validation, candidate-set retry, and one-shot repair. Our primary protocol is *non-oracle*: the repair stage never receives the benchmark gold tool identity, and missing tool names are handled by retrying over only the visible candidate tools. On Qwen3.5-9B, this protocol raises DICE-300 accuracy from 0.717 under drift to 0.813 after repair and BFCL-200 accuracy from 0.580 to 0.805, with zero repair harms on the originally-correct slice in both benchmarks. A naive-retry baseline reaches only 0.707 on DICE and 0.585 on BFCL, showing that the gains do not come from a second attempt alone. On our DICE-300 Qwen3.5-9B ablation, candidate-set drift dominates the base damage: description-only and schema-only drift cause < 1.5 point degradations on DICE, while distractor tools account for nearly all of the drop. In the non-oracle setting, missing tool names are a major bottleneck; adding a candidate-set retry reduces unresolved repair targets from 35 to 15 on DICE and from 34 to 9 on BFCL. Additional oracle-constrained sweeps across other models serve as upper bounds and error-analysis context, but we treat the non-oracle Qwen3.5-9B results as the primary deployment-facing evidence. A closed-model BFCL sanity check on GPT-4o-mini shows much smaller drift damage ($0.865 \rightarrow 0.840$) and a smaller, non-significant repair gain to 0.860 with zero harms, suggesting stronger closed models are less brittle but still compatible with the same repair layer.

Keywords: tool calling · language models · robustness · interface drift · repair

1 Introduction

Modern language-model agents increasingly depend on external tools for retrieval, execution, planning, and actuation. In practice, however, tool use is mediated entirely through text: tool descriptions, parameter schemas, enum values, and candidate tool lists. Recent work shows that these interface details strongly affect tool selection and argument generation even when the tool semantics are

unchanged [3, 1]. This creates a deployment problem: tool interfaces drift over time as developers rename parameters, revise descriptions, add examples, or expand candidate tool sets.

We study this problem as *interface drift*: benign, semantics-preserving changes to tool interfaces that occur after deployment. Our goal is not to train a new tool-calling model, but to quantify drift sensitivity and test whether a compact inference-time repair layer can recover reliability without retraining.

We instantiate drift with three deployment-motivated synthetic perturbations: (i) stale description examples that mention legacy field names, (ii) schema renaming, and (iii) distractor tools. Our primary evaluation uses Qwen3.5-9B on a 300-example DICE-BENCH slice [4] and a 200-example BFCL slice [12] under a deployment-realistic non-oracle repair protocol. In that protocol, validation runs against the visible tool selected by the model; if the model omits the tool name, a candidate-set retry over the visible tools resolves a concrete target before repair. We also retain earlier oracle-constrained sweeps on additional open models as upper-bound context and for secondary analyses.

This paper makes four empirical contributions that substantially revise the narrative of prior interface-drift work:

1. **Non-oracle inference-time repair recovers drift-induced failures with zero harms on the primary model.** On Qwen3.5-9B, SchemaShield-Lite raises DICE-300 accuracy from 0.717 to 0.813 and BFCL-200 accuracy from 0.580 to 0.805, while preserving zero repair harms on the originally-correct slice in both benchmarks.
2. **Naive retry does not explain the gains.** On DICE-300, naive retry reaches 0.707 versus repaired 0.813; on BFCL, 0.585 versus 0.805. A second attempt alone therefore cannot explain the recovery.
3. **Candidate-set drift dominates the damage on our primary-model ablation.** Under description-only drift or schema-only drift, Qwen3.5-9B accuracy drops by just 0.013 and 0.007 points respectively. Adding distractor tools causes a 0.047 drop, matching the full combined drift.
4. **Missing tool names are the main non-oracle bottleneck, and a candidate-set retry largely closes that gap.** In the non-oracle setting, adding a candidate-set retry improves repaired accuracy from 0.780 to 0.813 on DICE and from 0.725 to 0.805 on BFCL, while reducing unresolved repair targets from 35 to 15 and from 34 to 9 respectively.

2 Related Work

Recent work has established that tool interfaces are not neutral wrappers around tool use; they are part of the learning and inference problem itself. Guo et al. [3] show that rewriting tool descriptions can substantially improve agent reliability, especially when candidate tool sets are large. Faghih et al. [1] show the opposite direction: small edits to tool descriptions can dramatically shift tool preference, exposing a fragility in description-mediated tool use.

Several recent papers study robustness and recovery in tool-use systems. Zhang et al. [14] show that execution-error recovery remains a major weakness, particularly for smaller models, and propose RL-based training to improve it. Our setting is narrower and more deployment-oriented: we target inference-time robustness under interface drift rather than post-training optimization.

Benchmark work has also highlighted that realistic function calling is still far from solved. ComplexFuncBench introduces multi-step and constrained long-context function calling [15], while DICE-BENCH focuses on multi-round, multi-party dialogues with tool-related information dispersed across conversation turns [4]. We evaluate on both DICE-BENCH and the Berkeley Function Calling Leaderboard (BFCL) [12] because they capture complementary settings: DICE-BENCH exercises conversational multi-turn dialogues, while BFCL tests single-turn tool calls across diverse APIs.

Broader tool-use work has focused on scaling tool coverage and agent behavior rather than on post-deployment interface drift. ToolLLM and ToolBench study large-scale API-use training and evaluation across thousands of real-world tools [8], while Gorilla emphasizes retrieval and function selection over large API collections [7]. Agent frameworks such as ReAct and Reflexion show that multi-step reasoning, action, and self-correction can materially improve tool-using systems [13, 9]. Our setting is narrower: we isolate a deployment-time robustness problem in which tool semantics are fixed but interface surface form changes after release.

This setting also connects to software-engineering work on API evolution and contract checking. AutoGuard, for example, compares OpenAPI descriptions across versions and flags changes to methods, parameters, responses, and schemas [5]. We borrow the intuition that interface evolution is a first-class engineering concern, but study its effect on LLM tool use rather than on conventional API clients.

Compared with dataset-generation work such as APIGen [6], our objective is different. We do not create new large-scale tool-calling training data. Instead, we evaluate whether a lightweight repair layer can stabilize existing models against a specific class of post-deployment failures.

3 Problem Formulation

Let a tool be represented by an interface $I = (d, s, C)$, where d is the natural-language description, s is the argument schema, and C is the candidate tool set shown to the model. The underlying tool semantics are assumed fixed. An *interface drift* operator Δ maps I to a new interface $\Delta(I)$ that preserves tool semantics but perturbs surface form. For clarity, we view Δ as a composition of three operators,

$$\Delta = \delta_c \circ \delta_s \circ \delta_d,$$

where δ_d edits only the description text d, δ_s edits only the schema surface form s, and δ_c edits only the visible candidate set C. Each operator is semantics-preserving with respect to the benchmark gold tool call; the evaluation question

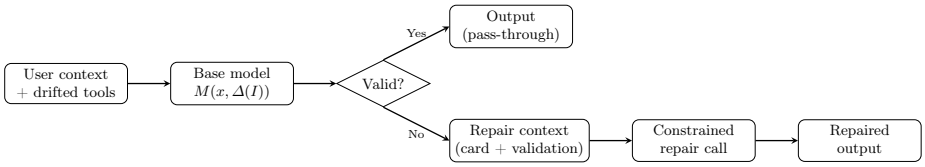

Fig. 1. SchemaShield-Lite pipeline. The base model receives user context and drifted tool definitions. If the emitted tool call fails validation, a repair call is issued with structured validation errors. In the primary non-oracle protocol, missing tool names are first resolved by retrying over the visible candidate tools; once a visible target tool is resolved, a repair step may be constrained to that selected tool where provider support allows.

is whether the model remains robust to these surface-form changes at inference time.

We study three drift types:

- **Description drift**: add stale or legacy examples to the description that refer to older field names.
- **Schema drift**: rename parameters while preserving semantics.
- **Candidate-set drift**: add distractor tools with overlapping descriptions.

Given user context x, tool interface I, and model M, the base system outputs a tool call $y = M(x, I)$. We evaluate three stages:

1. **Original**: inference on the original tool interface.
2. **Drifted**: inference on the drifted interface $\Delta(I)$.
3. **Repaired**: inference-time repair of an invalid or incomplete drifted call.

Our central question is whether an inference-time repair layer can recover the performance lost from Original to Drifted without harming examples that were already correct.

4 Method

4.1 SchemaShield-Lite

SchemaShield-Lite is a lightweight inference-time robustness layer with three parts. Figure 1 sketches the full pipeline.

Canonical tool card. Each tool is rendered into a compact canonical form containing the tool name, description, required fields, and field-level type information. This normalizes interface presentation across tools and emphasizes current field names.

Validation. After the base model emits a tool call, a validator checks tool name, required fields, and argument structure. Validation produces a structured error report rather than a binary failure signal.

One-shot repair. If the drifted call is invalid, the model receives a repair prompt containing the task context, canonical tool card, invalid call, and validation errors. In the primary non-oracle protocol, repair proceeds in two stages. If the base call names a visible tool, validation runs against that tool and a constrained repair call is issued for that same visible tool. If the base call omits the tool name, we first run a candidate-set retry over the visible tools to produce a concrete tool call; once a visible tool is resolved, a second repair call may be constrained to that selected tool. Where provider support allows it, this second step uses *forced-tool repair*; otherwise we fall back to unconstrained tool calling or JSON repair.

As our component ablation in Section 6.5 shows, it is the combination of structured validation errors and tool-constrained repair that does the heavy lifting once a visible target tool has been resolved. The canonical card is useful at the initial prediction step as drift-prevention prompting, but contributes negligibly to repair-stage recovery in our oracle-constrained upper-bound ablations.

4.2 Repair Prompting Details

Two implementation details turned out to matter materially:

- The repair prompt explicitly instructs the model to infer all required fields from the dialogue even when the invalid call is empty.
- When the invalid call omits the tool name, a candidate-set retry over the visible tools is needed before tool-constrained repair can fire.
- The repair stage uses a larger token budget than the base prediction stage. Raising repair capacity from the earlier default to 512 tokens fixed truncated repair outputs on longer DICE dialogues.

5 Experimental Setup

5.1 Benchmarks and Data Slices

We evaluate on two benchmarks. For DICE-BENCH [4], we use a 300-example subset drawn from single-function round-1 dialogues (allowing duplicate functions to reach sufficient scale). For BFCL [12], we use a 200-example subset drawn in equal parts from four categories: simple Python, multiple-function, live-simple, and live-multiple.

5.2 Models

We evaluate four open instruction-tuned models accessed via OpenRouter:

- **Qwen3.5-9B**: our primary model.
- **Qwen3.5-35B-A3B**: larger MoE variant.
- **Llama-4-Scout**: Meta's 2025 open model.
- **Mistral-Small-3.2-24B**: recent Mistral dense model.

All experiments use temperature 0.0 for deterministic-style evaluation.

5.3 Drift Configurations

For DICE main runs, we use *mild* severity: two description modes (`legacy_example`, `marketing`), two schema modes (`rename_parameters`, `reorder_parameters`), and one distractor tool. For BFCL we use *medium* severity: three description modes, three schema modes, and three distractors, reflecting a more aggressive drift scenario. These operators are author-designed and deployment-motivated rather than mined from real API changelogs; we return to that limitation in Section 8. Still, the change families do match public API-evolution practice. GitHub's versioned REST API explicitly treats removed or renamed parameters, added required parameters, removed response fields, and added operations as standard compatibility events [2]. Stripe changelog entries similarly replace older field groups with consolidated ones (for example, replacing Checkout Session line-item fields with `price`/`price_data`) and later add new optional parameters to existing endpoints [10, 11]. If each operation is wrapped as a tool, such additive API changes directly map to candidate-set growth. What remains less accessible from public changelogs is description/example drift: documentation-example edits are rarely versioned as cleanly as schema or operation changes, so our stale-example perturbation remains a heuristic rather than a mined distribution.

5.4 Repair Protocols

We use two repair protocols. Our *primary* protocol is non-oracle and is the main evidence in this paper: the validator checks the model-selected visible tool, and when the model omits the tool name, a candidate-set retry over only the visible tools resolves a target before repair. For broader model coverage and component ablations, we also retain earlier *oracle-constrained upper-bound* sweeps in which the repair stage is given the benchmark target tool. These upper-bound runs remain useful for cross-model context and ablation, but they are not the paper's deployment-facing evidence.

5.5 Metrics

We report:

- **Original score**: accuracy on the unmodified interface.
- **Drifted score**: accuracy under interface drift.
- **Repaired score**: accuracy after one-shot repair.
- **Naive retry score**: accuracy under a second identical drifted call (no canonical card, no validation feedback). This baseline isolates the value of simply retrying.
- **Originally-correct slice**: the subset of examples that the model solves correctly before drift. This slice isolates *true drift harm* from baseline errors.
- **Repair recoveries**: drift-induced failures recovered by repair.
- **Repair harms**: drifted-correct examples broken by repair.
- **Repair trigger rate**: fraction of examples for which the validator invokes the repair step.

Table 1. Primary non-oracle Qwen3.5-9B results on DICE-300 and BFCL-200, with oracle-constrained upper bounds for reference. Brackets give 95% bootstrap confidence intervals (seed 42). OC = originally-correct count; Rec = recoveries on drift-induced failures; Harms = repair harms on originally-correct slice; Trig% = repair trigger rate.

Benchmark	Protocol	Orig. [95% CI]	Drift. [95% CI]	Repair [95% CI]	OC	Rec/Miss	Harms	Trig%
DICE-300	Non-oracle + candidate retry	0.757 [0.70, 0.80]	0.717 [0.67, 0.77]	**0.813 [0.77, 0.86]**	227	19/22	0	16.0
	Oracle upper bound	0.750 [0.70, 0.80]	0.713 [0.66, 0.76]	**0.827 [0.79, 0.87]**	225	17/18	0	21.0
BFCL-200	Non-oracle + candidate retry	0.790 [0.74, 0.84]	0.580 [0.51, 0.65]	**0.805 [0.75, 0.86]**	158	33/45	0	32.0
	Oracle upper bound	0.790 [0.74, 0.84]	0.640 [0.58, 0.71]	**0.795 [0.74, 0.85]**	158	23/35	0	31.0

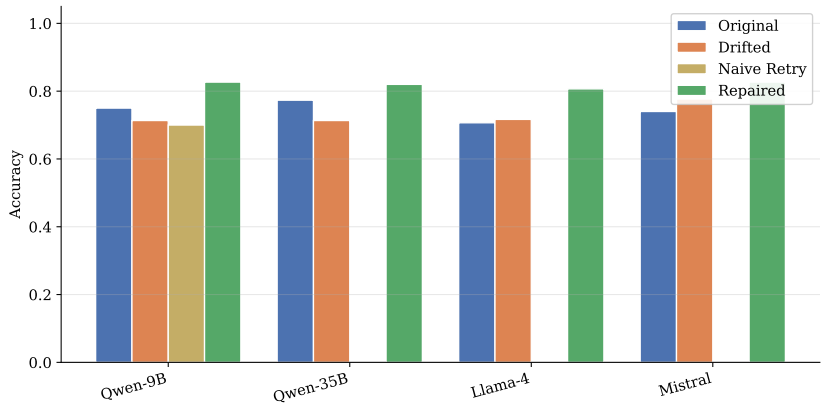

Fig. 2. Oracle-constrained upper-bound DICE sweep across the four evaluated models. We retain these runs as secondary cross-model context; the paper's primary deployment-facing evidence is the non-oracle Qwen3.5-9B protocol in Table 1.

6 Results

6.1 Primary Non-Oracle Results and Oracle Upper Bounds

Table 1 summarizes our primary non-oracle Qwen3.5-9B results on DICE-300 and BFCL-200, together with earlier oracle-constrained upper bounds for reference. Under the non-oracle candidate-retry protocol, SchemaShield-Lite improves over the drifted baseline on both benchmarks with *zero repair harms* on the originally-correct slice.

The primary non-oracle results are strong on both benchmarks. On DICE-300, repair improves Qwen3.5-9B from 0.717 to 0.813 and recovers 19/22 drift-induced failures on the originally-correct slice. On BFCL-200, repair improves from 0.580 to 0.805 and recovers 33/45 drift-induced failures. The remaining gap to the oracle upper bound is small on DICE (0.813 vs 0.827), and on BFCL the final non-oracle candidate-retry run slightly exceeds the earlier oracle-style run (0.805 vs 0.795).

We therefore focus our interpretation on the large repaired-vs-drifted gaps, not on small differences between nearby runs. The bootstrap intervals in Table 1 support the main claim that structured repair helps, but they do not justify

Table 2. Naive retry vs the primary non-oracle SchemaShield-Lite protocol on Qwen3.5-9B.

Benchmark	Drifted	Naive Retry	SchemaShield-Lite	Δ over Drifted
DICE-300	0.717	0.707	**0.813**	+0.097
BFCL-200	0.580	0.585	**0.805**	+0.225

strong claims about minor oracle-vs-non-oracle differences. Paired exact McNemar tests on the final non-oracle runs also support the main repair effect: repaired beats drifted with $p = 3.7 \times 10^{-9}$ on DICE and $p = 5.7 \times 10^{-14}$ on BFCL. On shared examples, the repaired-score difference between the oracle upper bound and the final non-oracle protocol is itself not significant on either benchmark (DICE exact McNemar $p = 0.125$; BFCL $p = 0.754$), which is consistent with the small raw score gaps in Table 1.

The candidate-set retry is load-bearing in the non-oracle setting. Relative to an earlier non-oracle run that simply skipped unresolved tool names, candidate retry raises repaired accuracy from 0.780 to 0.813 on DICE and from 0.725 to 0.805 on BFCL, while reducing unresolved repair targets from 35 to 15 and from 34 to 9 respectively.

Repair cost overhead. On the final non-oracle Qwen3.5-9B runs, SchemaShield-Lite triggers repair on 48/300 examples (16.0%) in DICE and 64/200 examples (32.0%) in BFCL. Total repair tokens are 103,221 against 316,185 drifted tokens on DICE (32.6% relative overhead) and 154,249 against 327,289 drifted tokens on BFCL (47.1% relative overhead). Because repair is gated by validation, this overhead is paid only when the drifted call is invalid or incomplete. On a refreshed BFCL-200 live rerun of the same non-oracle protocol with latency instrumentation, average original-call latency is 8.8s, average drifted-call latency is 9.3s, and average repair-call latency when repair fires is 13.0s. With a 27.0% repair trigger rate on that rerun, this corresponds to roughly 3.5s expected added latency per example.

6.2 Naive Retry Does Not Explain the Gains

A first-order alternative explanation is that repair works simply because the model gets a second attempt. We test this by running a *naive retry* call that uses the same drifted tools but without the canonical card or the validation error report. Table 2 shows that naive retry remains far below structured repair: 0.707 vs 0.813 on DICE and 0.585 vs 0.805 on BFCL. Naive retry is therefore insufficient to explain the gains, even though it is nearly flat with drifted performance on BFCL. The paired repaired-vs-naive contrast is also highly significant on both benchmarks (exact McNemar $p = 4.1 \times 10^{-9}$ on DICE; $p = 1.1 \times 10^{-13}$ on BFCL).

Table 3. Decomposition of repaired improvements on the primary non-oracle Qwen3.5-9B runs.

Benchmark	Total gains	Drift recovery	Baseline patching	Harms
DICE-300	29	19	10	0
BFCL-200	45	33	12	0

Table 4. Drift-type ablation on DICE-300 with Qwen3.5-9B. Orig./Drift. scores come from the base model; Repair scores are from the oracle-constrained upper-bound repair protocol.

Drift type	Orig.	Drift.	Repair	Drift Drop
Description only	0.760	0.747	0.813	−0.013
Schema only	0.767	0.760	0.830	−0.007
Candidate-set only	0.757	0.710	0.830	−0.047
All combined	0.763	0.713	0.823	−0.050

6.3 Repair Gains Are Mostly Drift Recovery, Not Only Baseline Patching

The "repair > original" effect is real, but it does not dominate the main story. Table 3 decomposes every repaired improvement over the drifted call into either *drift recovery* on the originally-correct slice or *baseline patching* on examples the base model already missed before drift. Most repaired gains still come from genuine drift recovery: 19/29 on DICE and 33/45 on BFCL. Baseline patching is material, especially on BFCL, and should be read as a secondary benefit rather than the paper's central causal claim.

6.4 Drift-Type Ablation: Candidate-Set Drift Dominates

Table 4 decomposes the drift pipeline into its three components and evaluates each in isolation at DICE-300 on Qwen3.5-9B. Description drift alone causes only a 0.013 point accuracy drop. Schema drift alone causes only 0.007. By contrast, candidate-set drift (adding a distractor tool) causes a 0.047 drop—matching the full combined drift scenario (0.050). Because this ablation is about where the damage comes from, the main interpretation rests on the Original-to-Drifted delta rather than the repair column.

This result qualifies a common framing in the interface-drift literature. On this Qwen3.5-9B DICE-300 ablation, minor description rewording and semantics-preserving parameter renames do not meaningfully degrade accuracy—the model is evidently robust to this class of surface perturbation. The observed failure mode is *candidate-set confusion*: when a distractor tool with an overlapping description is added to the candidate set, tool selection degrades. This has implications for deployment-time monitoring: teams should prioritize tracking tool-catalog growth over monitoring description or schema revisions.

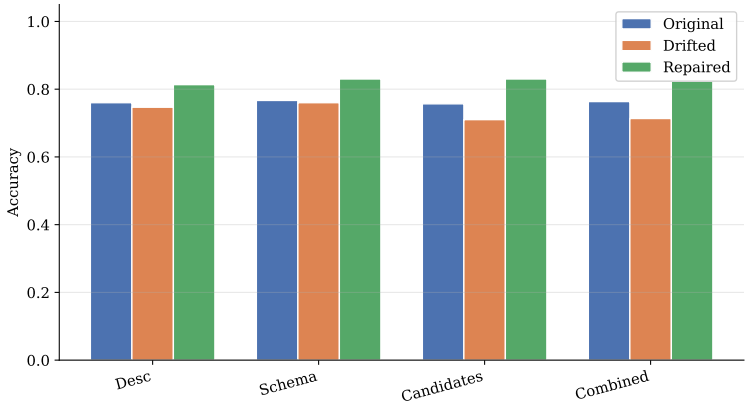

Fig. 3. Drift-type ablation on DICE-300 with Qwen3.5-9B. Description-only and schema-only drift cause negligible accuracy drops. Adding distractor tools reproduces essentially all of the damage of the full combined drift.

Table 5. Oracle-constrained upper-bound component ablation on DICE-300 and BFCL-200 with Qwen3.5-9B. BFCL uses medium-severity drift, DICE uses mild.

Benchmark	Variant	Drift.	Repair	Rec/Miss	Harms	Trig%
DICE-300	Canonical card only	0.770	0.770	0/9	0	0.0
	Validation + retry (no card)	0.730	0.830	16/17	0	20.0
	Full SchemaShield-Lite	0.713	**0.827**	17/18	0	21.0
BFCL-200	Canonical card only	0.810	0.810	0/11	0	0.0
	Validation + retry (no card)	0.645	**0.820**	26/34	0	29.5
	Full SchemaShield-Lite	0.640	0.795	23/35	0	31.0

6.5 Upper-Bound Component Ablation: Validation, Not Canonical Card, Drives Recovery

To isolate which parts of the repair stack matter once a target tool has already been resolved, Table 5 evaluates three variants of SchemaShield-Lite in an oracle-constrained upper-bound setting on both DICE-300 and BFCL-200 with Qwen3.5-9B: (i) canonical card only, where the card is prepended to the initial prompt but no validation/repair is applied; (ii) validation-plus-retry, which uses validation errors in a repair prompt but omits the canonical tool card from the repair context; and (iii) full SchemaShield-Lite (card + validation + repair).

Within this oracle-constrained setting, the BFCL rows confirm and *strengthen* the DICE finding in three ways. First, the canonical card at initial-prediction time provides even stronger drift prevention on BFCL than on DICE: the base drifted score of 0.640 jumps to 0.810—a 17.0-point drift-prevention effect. Second, the card-only configuration still never triggers the validator and recovers zero drift-induced failures, because its valid-but-imperfect outputs pass schema

checking without entering the repair path. Third, validation-plus-retry without the canonical card at repair time reaches a *higher* repaired accuracy on BFCL than the full pipeline (0.820 vs 0.795, and 26/34 recoveries vs 23/35), matching the pattern observed on DICE where validation-only nearly ties the full pipeline.

Taken together across both benchmarks, these upper-bound numbers localize the load-bearing component of SchemaShield-Lite to the validation error feedback loop combined with tool-constrained repair, not the canonical tool card presented at repair time. The card earns its keep as *drift-prevention prompting* at the initial prediction step—where BFCL shows it can dominate the remaining drift damage—but can be dropped from the repair prompt without hurting recovery once the target tool has already been resolved.

6.6 Error Analysis

Across the oracle-constrained DICE upper-bound sweeps, all repair harms are zero. However, not every drift miss is recovered. On Qwen3.5-35B, repair recovers 20 of 25 drift misses; on Llama-4-Scout, 12 of 16; on Mistral, 8 of 12. We manually inspected every unrecovered case on these three runs and classified them against the gold call.

The pattern is consistent across models: nearly all unrecovered examples involve the repair call selecting the correct tool with well-formed arguments whose string values disagree with gold only by formatting or paraphrase. On Qwen3.5-35B (5 unrecovered), these include one strict whitespace/punctuation difference ("`the quick brown fox...`" vs "`The quick brown fox....`") and three enum-format or paraphrase-level mismatches (e.g. `regular_season` vs `regular season`, `the office` vs `Office`); a single genuine failure has an empty arguments object after repair. On Llama-4-Scout (4 unrecovered), all four are format-level: `ISO`-prefixed dates vs two-field dates (e.g. `2023-11-10` vs `11-10`), or JSON with quoted-number string values rather than numeric literals. On Mistral-24B (4 unrecovered), the misses include near-synonym topics (`inspirational` vs `inspiration`, `AI` vs `artificial intelligence`, `Google` vs `GOOGL`) and one wrong numeric amount. Across 13 unrecovered cases in total, only 2 are genuine inference failures (empty arguments and one wrong value); the remaining 11 are strict-match artifacts rather than true drift failures. The practical implication is that the paper's reported recovery rates are a lower bound: a lenient evaluator would count most of the unrecovered cases as correct.

On the primary non-oracle Qwen3.5-9B runs, repair improves over the drifted call on 29 DICE examples and 45 BFCL examples in total. Not all of these are pure drift recovery: 10 of the DICE gains and 12 of the BFCL gains are baseline-error patching on examples the base model already missed before drift. This side effect is useful in deployment, but it reinforces the need to separate drift recovery from general post-hoc correction in the paper's reporting.

Mistral-24B exhibits an unusual pattern: its drifted score (0.777) is slightly higher than its original score (0.740). Inspection of the 23 examples where Mistral's original call misses but the drifted call matches shows that almost none of them involve the model recovering extra information from the drifted

Table 6. Compact summary of unrecovered oracle-constrained DICE cases on the three non-primary models. "Artifact" includes formatting/paraphrase mismatches that appear semantically correct under manual inspection.

Model	Unrecovered	Artifact-like	Genuine failure
Qwen3.5-35B-A3B	5	4	1
Llama-4-Scout	4	4	0
Mistral-24B	4	3	1
Total	13	11	2

description. Instead, 15/23 are strict-match artifacts on the original call: formatting differences in the value (`apple` vs `apples`, `bread, eggs, cheese` vs `bread,eggs,cheese`, `https://wikipedia.org/` vs `https://wikipedia.org`), and only a handful reflect the model missing a required field in the original call (4/23) or choosing a wrong tool name (4/23). The drifted call happens to land in the tokenization bucket that the scorer accepts, pushing the scored accuracy slightly above the original. We therefore read the Mistral "drift > original" effect as scorer noise rather than as evidence that description drift helps Mistral, and revise the original speculation accordingly.

7 Discussion

Our results support three narrower conclusions than the original SchemaShield-Lite framing suggested. First, non-oracle inference-time repair is a viable robustness layer for tool-calling systems under interface drift: on Qwen3.5-9B it improves over drifted performance on both DICE and BFCL while preserving zero harms on the originally-correct slice. Second, the main non-oracle bottleneck is missing tool names; a candidate-set retry over the visible tools closes much of the gap to the oracle upper bound. Third, the *target* of robustness work is candidate-set drift rather than description or schema drift.

Together these findings simplify the deployment picture. A practitioner does not need a heavy per-tool canonicalization pipeline; a visible-tool validator, a candidate-set retry for missing tool names, and a constrained repair call once a tool is resolved already recover most of the damage we observe. And the monitoring story is clearer: the first-order cause of post-deployment tool-calling regressions is likely tool-catalog expansion, not description or parameter rename churn.

These same results also matter for adversarial interface manipulation. Faghih et al. [1] show that small description edits can steer tool selection even when semantics are unchanged. Our ablation suggests that the more serious surface to monitor is not just description wording in isolation, but the combination of description overlap and candidate-set expansion: once distractor tools enter the visible set, routing degrades quickly. In that sense, the non-oracle validator-plus-

repair stack is relevant not only to benign interface drift but also to description-surface attacks that aim to induce tool confusion.

7.1 Second Non-Oracle Replication: Llama-4-Scout on BFCL

To reduce the single-model dependence of the primary story, we also ran the same BFCL-200 non-oracle candidate-retry protocol on Llama-4-Scout. Repair still improves over the drifted call ($0.635 \rightarrow 0.665$, exact McNemar $p = 0.031$) and again produces zero repair harms on the originally-correct slice. However, this run exposes an important portability failure mode: the model omits the tool name frequently, leaving 43 unresolved repair targets and limiting recovery on the originally-correct slice to just 1/11 drift misses. The run therefore supports some portability of the repair effect, but also reinforces that non-oracle repair quality depends heavily on the base model's ability to emit a usable tool name. We keep Qwen3.5-9B as the clean primary evidence and treat this Llama result as a secondary replication plus failure-mode analysis.

7.2 Closed-Model Sanity Check: GPT-4o-mini on BFCL

Finally, we ran a BFCL-200 non-oracle sanity check on GPT-4o-mini after adding provider-safe tool-name aliasing for OpenAI-style tool schemas. On this stronger closed model, drift damage is much smaller to begin with: accuracy falls only from 0.865 to 0.840. Repair lifts the score to 0.860 with zero harms and zero unresolved repair targets, but the gain is small and not significant on this 200-example slice (exact McNemar $p = 0.125$). We therefore read the GPT-4o-mini result as a useful credibility check rather than as a new headline number: the problem appears less severe on a stronger closed model, yet the same non-oracle repair stack does not appear to hurt it.

8 Limitations

Several limitations remain. First, although we now include a second non-oracle BFCL replication on Llama-4-Scout, the cleanest deployment-facing evidence still comes from a single primary model, Qwen3.5-9B; the broader multi-model results retained in the paper remain mostly oracle-constrained upper bounds rather than full non-oracle replications. Second, the BFCL evaluation still covers only two successful open-model non-oracle replications, and one of them is materially bottlenecked by missing tool names; we also include one closed-model BFCL sanity check, but not yet a broader closed-model sweep. Third, our drift operators focus on stale descriptions, renamed fields, and distractors; other forms of interface drift (for example, type narrowing, enum replacement, or multi-tool refactors) remain untested, and the perturbations are author-designed rather than mined from real API evolution logs. Fourth, evaluation uses hosted inference and so may incorporate provider-specific tool-calling behavior. Fifth, the

"repair > original" effect indicates that SchemaShield-Lite is also patching baseline failures—useful in deployment, but a complication for clean attribution to drift recovery specifically. Sixth, strict string-matching against gold calls undercounts true recoveries: manual inspection of the 13 unrecovered cases on the three non-primary models (Section 6.6) shows that 11 are formatting or paraphrase artifacts, so the reported recovery rates should be read as lower bounds. Finally, while we now report broader cross-protocol significance for oracle-vs-non-oracle repaired outcomes on the shared Qwen slices, we still lack a broader wall-clock latency sweep across benchmarks, providers, and models.

9 Conclusion

We studied SchemaShield-Lite, a lightweight inference-time validation-and-repair stack for tool calling under interface drift. On Qwen3.5-9B under a non-oracle candidate-retry protocol, SchemaShield-Lite improves DICE-300 from 0.717 to 0.813 and BFCL-200 from 0.580 to 0.805, with zero repair harms on the originally-correct slice in both benchmarks. Three findings qualify the picture. A naive-retry baseline does not explain the gains. Description and schema drift cause only small degradations on our DICE slice, while candidate-set drift is the dominant failure mode. And in the non-oracle setting, missing tool names are the main repair bottleneck: a candidate-set retry over the visible tools substantially reduces unresolved cases and closes most of the gap to oracle-constrained upper bounds. Additional oracle-constrained sweeps across other models and upper-bound ablations remain useful context, and a BFCL sanity check on GPT-4o-mini suggests that stronger closed models suffer less drift damage to begin with. Still, the paper's main evidence is the non-oracle deployment-facing Qwen protocol.

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